

1 Analysis 3: Is Spatial Evidence Present, and Can a Frozen Model Use It?

Motivation and hypotheses. Idea 5 adds explicit geometry and body/hand pose for norm reasoning. We test two of its premises without training: H1, spatial content is concentrated in a subset of norm categories and is recoverable from language; H2, appending pose, segmentation, or depth maps changes a frozen VLM’s answers beyond extra-image and mismatched-map controls.

Data and protocol. Language. For all 1,853 EgoNormia items [?] we embed each action–justification option with mean-pooled bert-base-uncased and compare against Empath categories and a fixed list of spatial terms (distance, space, behind, aside, ...). Probes are logistic regressions with folds grouped by source video. Vision. On 40 items from distinct videos, fixed by hash before inference, we derive pose (MediaPipe), segmentation (YOLOv8n-seg), and relative depth (Depth Anything V2) from five pre-action frames. Frozen models answer from RGB alone or RGB plus maps; controls repeat the RGB grid or use maps from another video. Scores require action and justification both correct.

Results. H1. A spatial term appears in 24.6% of correct options, most often for Privacy (40.0%) and Proxemics (33.7%) and least for Cooperation (20.9%; Figure 1). Proxemics membership is predicted with AUC 0.67 from BERT, against 0.58 from Empath and 0.59 from the lexicon; a t-SNE shows overlapping regions rather than clusters (Appendix). As in Analysis 1, option text alone identifies the answer on 58.6% of items (chance 25%). H2. Maps help no model. Qwen3-VL-8B scores 16/40 on RGB, 15/40 with aligned pose, 15/40 with pose from an unrelated video, and 13/40 with RGB repeated. Claude Opus 5.5 scores 12/15 on RGB and 12–13/15 in every other condition, giving identical answers across all 16 conditions on 14 items; the one item that changes also changes under both controls. A GPT-5.6 agent scores 11/15 on RGB, and none of its four failures is recovered by any map. Pose or hands are detected in only 40.5% of frames, and segmentation mislabels objects (horse, bird on a construction site).

Discussion of Analysis 3. Idea 5’s premises hold only in part. Spatial content is a minority signal: three in four correct answers use no spatial vocabulary, and it concentrates in Proxemics and Privacy. Appending geometry to a frozen model does nothing mea-

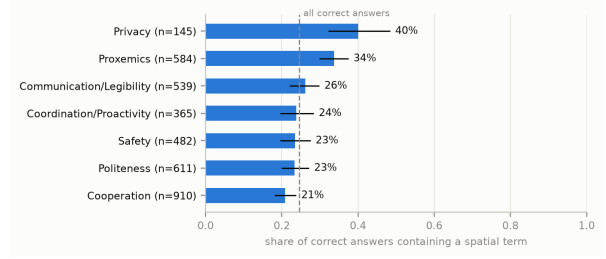


Figure 1: Share of correct options containing a spatial term, by norm category (categories overlap). Bars: 95% bootstrap intervals; dashed line: all correct options.

surable, and the mismatched-map control shows why: models do not read the maps. The maps are derived from the same RGB frames, so they re-encode rather than add information, and they are empty in most egocentric frames. With 15–40 items and agent-based runs for GPT and Claude, this shows no immediate benefit, not that adaptation would fail.

END OF ANALYSIS